# Hariprasad AP

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## Professional Summary

Final-year MCA student building offline-first, evaluation-driven ML systems - a custom CNN that detects pneumonia from chest X-rays at 90.4% accuracy and 96.8% recall, and a fully local semantic search engine using FAISS and Sentence Transformers. Strong in Python, model evaluation, and shipping working software on constrained hardware. Seeking an AI/ML Engineering role.

## Projects

### Pneumonia Detection CNN - Chest X-Ray Classifier — Live Demo — GitHub

*Python, TensorFlow/Keras, Scikit-learn, Streamlit*

- Designed and trained a custom CNN from scratch (no transfer learning) to classify chest X-rays as Normal or Pneumonia, achieving **90.38% accuracy** and **96.84% recall** on a 624-image held-out test set.
- Handled severe class imbalance (74% pneumonia vs 26% normal) using dynamic class weighting, reducing missed pneumonia cases to only **12 false negatives** out of 390 positive cases.
- Built an isolated evaluation pipeline with confusion matrix and ROC-AUC analysis (**0.954 AUC**) to validate model reliability before deployment.
- Deployed a 4-page Streamlit dashboard for live inference, performance analytics, and technical model inspection.

### Local AI File Search Assistant — GitHub

*Python, PySide6, Sentence Transformers, FAISS, SQLite (FTS5), Gemini API*

- Built a hybrid search engine combining SQLite FTS5 keyword search with FAISS semantic search ($0.7 \times$ semantic $+ 0.3 \times$ BM25), returning ranked results in under 3 seconds across 500+ indexed files.
- Implemented dense vector embeddings using Sentence Transformers (`all-MiniLM-L6-v2`) over 800-character overlapping text chunks, indexed with FAISS for chunk-level semantic retrieval instead of whole-file matching.
- Built a Retrieval-Augmented Generation (RAG) chat feature that retrieves the top-5 most relevant chunks per query and passes them to the Gemini API for multi-turn conversational Q&A over indexed documents.
- Designed a thread-safe PySide6 desktop UI running indexing and embedding generation on background threads, keeping search fully on-device while the optional chat feature connects to Gemini.

### Scrappy AI - LLM-Powered Web Scraper — GitHub

*Python, Selenium, BeautifulSoup, Streamlit, Gemini API*

- Built an AI-powered scraper combining Selenium and BeautifulSoup extraction with Gemini API for natural-language-driven parsing of unstructured multi-page web data.
- Benchmarked local LLM inference (DeepSeek via Ollama) against Gemini API, identified hardware-constrained latency tradeoffs, and migrated to cloud inference for reliable performance.
- Built a Streamlit interface for scrape configuration, content cleaning, and structured result visualization.

## Technical Skills

**Languages:** Python, SQL; familiar with C/C++, Java
**AI/ML:** TensorFlow, Keras, CNNs, scikit-learn, Sentence Transformers, FAISS, NLP, LLM APIs (Gemini, Ollama)
**Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Data Preprocessing
**Frameworks & Tools:** Streamlit, Selenium, BeautifulSoup, Git, GitHub, Linux/Unix
**Databases & APIs:** SQLite, MySQL, REST APIs

## Education

**Presidency College (Autonomous)** — Bengaluru, India
*Master of Computer Applications (MCA)* — *Nov 2024 – Aug 2026*

**College of Applied Science IHRD** — Calicut, India
*Bachelor of Computer Applications (BCA)* — *Jul 2021 – Apr 2024*

## Certifications

- Deep Learning with Python: CNN, ANN & RNN – Coursera (Verify)
- AI Fluency: Framework & Foundations – Anthropic (Verify)
- Deloitte – Data Analytics Job Simulation (Forage) (Verify)
- Scientific Computing with Python – freecodecamp (Verify)